Properties of Partition Table

Byte Range	Description	Essential
0-445	Boot Code	No
446-461	Partition Entry 1	Yes
462-477	Partition Entry 2	Yes
478-493	Partition Entry 3	Yes
494-509	Partition Entry 4	Yes
510-511	Signature (0xAA55)	No

Dos Partition Table

Byte Range	Description	Essential
0-0	Bootable Flag	No (2)
1-3	Start CHS (8-bit head), (10-bit Cylinder), (6-bit sector)	Yes
4-4	Partition Type	No (1)
5-7	End CHS	Yes
8-11	Start LBA	Yes
12-15	Size in Sector	Yes

Dos Partition Entries

(1) The type value can be interpreted differently by different OS. Window will not mount a file system in a partition that does not support it. Linux on the other hand ignores this flag (Partition Type). It can put FAT file system into a partition NTFS.

(2) The flag indicates which partition is a bootable one. With one OS the flag would be 0x80. With multiple OS, the boot code will prompt the users on which partition to boot from, the flag is not required.

Type (Hex)	Description
00	Empty
01	FAT12, CHS
04	FAT16, 16-32 MB, CHS
05	Microsoft Extended, CHS
06	FAT16, 32MB-2GB, CHS
07	NTFS
0b	FAT32, CHS
0c	FAT32, LBA
0e	FAT16, 32MB-3GB
0f	Microsoft Extended, LBA
11	Hidden FAT12, CHS
14	Hidden FAT16, 16-32MB, CHS
16	Hidden FAT16, 32MB-2GB, CHS
1b	Hidden FAT32, CHS
1c	Hidden FAT32, LBA
1e	Hidden FAT16, 32MB-2GB, LBA
42	Microsoft MBR, Dynamic Disk
81	Solaris x86
82	Linux Swap
83	Linux
84	Hibernation
85	Linux Extended
86	NTFS Volume Set
87	NTFS Volume Set
A0	Hibernation
A1	Hibernation
A5	FreeBSD
A6	OpenBSD
A8	Mac OSX
A9	NetBSD
Ab	Mac OSX Boot
B7	BSDI
B8	BSDI Swap
Ee	EFI GPT Disk
Ef	EFI System Partition
Fb	VmWare File System
fc	VmWare Swap

Some type values for DOS partitions

http://www.win.tue.nl/~aeb/partitions/partition_types-1.html

ID Name
00 Empty

To be precise: this is not used to designate unused area on the disk, but marks an unused partition table entry. (All other fields should be zero as well.) Unused area is not designated. Plan9 assumes that it can use everything not claimed for other systems in the partition table.

01 DOS 12-bit FAT

DOS is a family of single-user operating systems for PCs. 86-DOS (`QDOS' - Quick and Dirty OS) was a CP/M-like operating system written by Tim Paterson of Seattle Computer Products (1979). Microsoft bought it, renamed it to MS-DOS 1.0 and sold it to IBM (1980) to be delivered together with the first IBM PCs (1981). MS-DOS 2.0 (1983) was rather different, and designed to be somewhat Unix-like. It supported a hard disk (up to 16MB; up to 32MB for version 2.1). Version 3.3+ added the concept of partitions, where each partition is at most 32MB. (Compaq DOS 3.31 relaxed this restriction.) Since version 4.0 partitions can be 512 MB. Version 5.0 supports partitions up to 2 GB. Several clones exist: DR-DOS (from Digital Research, later part of Novell and called NovellDOS or NDOS, then owned by Caldera and called OpenDOS, then by its subsidiary Lineo who named it back to DR-DOS. See http://www.drdos.com/), PC-DOS (from IBM), FreeDOS, ... See Types of DOS. See comp.os.msdos.* and MSDOS partitioning summary. The type **01** is for partitions smaller than 16 MB.

02 XENIX root
03 XENIX /usr

Xenix is an old port of Unix V7. Microsoft Xenix OS was announced August 1980, a portable and commercial version of the Unix operating system for the Intel 8086, Zilog Z8000, Motorola M68000 and Digital Equipment PDP-11. Microsoft introduces XENIX 3.0 in April 1983. (Timeline of Microcomputers) SCO delivered its first Xenix for 8088/8086 in 1983. See comp.unix.xenix.sco.

04 DOS 3.0+ 16-bit FAT (up to 32M)

Matthias Paul writes: Some old DOS versions have had a bug which requires this partition to be located in the 1st physical 32 MB of the hard disk, hence for compatibility with these old issues, partitions located elsewhere should better be assigned the ID FAT16B (06h).

05 DOS 3.3+ Extended Partition

Supports at most 8.4 GB disks: with type **05** DOS/Windows will not use the extended BIOS call, even if it is available. See type **0f** below. Using type **05** for extended partitions beyond 8 GB may lead to data corruption with MSDOS.

An extended partition is a box containing a linked list of logical partitions. This chain (linked list) can have arbitrary length, but some FDISK versions refuse to make more logical partitions than there are drive letters available (e.g. MS-DOS

LASTDRIVE=26 is good for at most 24 disk partitions; Novell DOS 7+ allows LASTDRIVE=32).

06 DOS 3.31+ 16-bit FAT (over 32M)

Partitions, or at least the FAT16 filesystems created on them, are at most 2 GB for DOS and Windows 95/98 (at most 65536 clusters, each at most 32 KB). Windows NT can create up to 4 GB FAT16 filesystems (using 64 KB clusters), but these cause problems for DOS and Windows 95/98. Note that VFAT is 16-bit FAT with long filenames; FAT32 is a different filesystem.

07 OS/2 IFS (e.g., HPFS)

IFS = Installable File System. The best known example is HPFS. OS/2 will only look at partitions with ID 7 for any installed IFS (that's why the EXT2.IFS packet includes a special "Linux partition filter" device driver to fool OS/2 into thinking Linux partitions have ID 07). (Kai Henningsen (`kai@khms.westfalen.de`))

07 Windows NT NTFS

Filesystem introduced in Windows NT 3.1. It is rumoured that the Windows NT boot partition must be primary, and within the first 2 GB of the disk.

07 exFAT

Extended FAT, a.k.a. FAT64. Available in Microsoft Windows since CE 6.0 and Vista SP1. Allows 32 MB clusters and very large disks and files.

07 Advanced Unix
07 QNX2.x pre-1988 (see below under IDs 4d-4f)
08 OS/2 (v1.0-1.3 only)
08 AIX boot partition
08 SplitDrive
08 Commodore DOS

Matthias Paul writes: "This indicates a Commodore MS-DOS 3.x logically sectored FAT partition."

08 DELL partition spanning multiple drives
08 QNX 1.x and 2.x ("qny")

(according to QNX Partitions)

09 AIX data partition

Some reports interchange AIX boot & data. AIX is IBM's version of Unix. See comp.unix.aix.

09 Coherent filesystem

Coherent was a UNIX-type OS for the 286-386-486, marketed by Mark Williams Company led by Bob Swartz, renowned for its good documentation. It was introduced in 1980 and died 1 Feb 1995. The last versions are V3.2 for 286-386-486 and V4.0 (May 1992, using protected mode) for 386-486 only. It sold for $99 a copy, and the FAQ says that 40000 copies have been sold. See comp.os.coherent and this page. A Coherent partition has to be primary.

09 QNX 1.x and 2.x ("qnz")

(according to QNX Partitions)

0a OS/2 Boot Manager

OS/2 is the operating system designed by Microsoft and IBM to be the successor of MS-DOS. Dropped by Microsoft. See comp.os.os2. Windows 2000 actively tries to destroy OS/2 Boot Manager. See below.

0a Coherent swap partition
0a OPUS

Open Parallel Unisys Server. See Unisys.

0b WIN95 OSR2 FAT32

Partitions up to 2047GB. See Partition Types

0c WIN95 OSR2 FAT32, LBA-mapped

Extended-INT13 equivalent of **0b**.

0e WIN95: DOS 16-bit FAT, LBA-mapped
0f WIN95: Extended partition, LBA-mapped

Windows 95 uses **0e** and **0f** as the extended-INT13 equivalents of **06** and **05**. For the problems this causes, see Possible data loss with LBA and INT13 extensions. (Especially when going back and forth between MSDOS and Windows 95, strange things may happen with a type **0e** or **0f** partition.) Windows NT does not recognize the four W95 types **0b**, **0c**, **0e**, **0f** (Win95 Partition Types Not Recognized by Windows NT). DRDOS 7.03 does not support this type (but DRDOS 7.04 does).

10 OPUS (?)

Maybe decimal, for type **0a**.

11 Hidden DOS 12-bit FAT

When it boots a DOS partition, OS/2 Boot Manager will hide all primary DOS partitions except the one that is booted, by changing its ID: **01**, **04**, **06** becomes **11**, **14**, **16**. Also **07** becomes **17**.

11 Leading Edge DOS 3.x logically sectored FAT

(According to Matthias Paul.)

12 Configuration/diagnostics partition

ID 12 (decimal 18) is used by Compaq for their configuration utility partition. It is a FAT-compatible partition (about 6 MB) that boots into their utilities, and can be added to a LILO menu as if it were MS-DOS. (David C. Niemi) Stephen Collins reports a 12 MB partition with ID 12 on a Compaq 7330T. Tigran A. Aivazian reports a 40 MB partition with ID 12 on a 64 MB Compaq Proliant 1600. ID 12 is used by the Compaq Contura to denote its hibernation partition. (dan@fch.wimsey.bc.ca)

NCR has used ID 0x12 MS-DOS partitions for diagnostics and firmware support on their WorldMark systems since the mid-90s. DataLight's ROM-DOS has replaced MS-DOS on more recent systems. Partition sizes were once 72M (MS-DOS) but are now 40M (ROM-DOS).

Intel has begun offering ROM-DOS based "Service Partition" support on many OEM systems. This support initially used ID 0x98 but has recently changed to ID 0x12. Intel provides their own support for this partition in the form of a System Resource CD. Partition size has remained constant at 40M. See e.g. sds2.pdf. (Chuck Rouillard)

IBM also uses 0x12 for its Rescue and Recovery partition on Thinkpad laptops. See also thinkwiki.org.

14 Hidden DOS 16-bit FAT <32M

(Ralf Brown's interrupt list adds: `ID 14 resulted from using Novell DOS 7.0 FDISK to delete Linux Native partition')

14 AST DOS with logically sectored FAT

AST MS-DOS 3.x was an OEM version supporting 8 instead of the usual 4 partition entries in the MBR. These special MBRs can be detected by another signature in the MBR stored in front of the partition table.

16 Hidden DOS 16-bit FAT >=32M
17 Hidden IFS (e.g., HPFS)
18 AST SmartSleep Partition

Ascentia laptops have a `Zero Volt Suspend Partition' or `SmartSleep Partition' of size 2MB+memory size. See AST. Ralf Brown calls this the "AST Windows swapfile".

19 Unused

Claimed for Willowtech Photon coS (completely optimized system) by Willow Schlanger `willow@dezine.net`. See dejanews.

1b Hidden WIN95 OSR2 FAT32

1c Hidden WIN95 OSR2 FAT32, LBA-mapped

Sometimes a hidden IBM rescue partition.

1e Hidden WIN95 16-bit FAT, LBA-mapped
20 Unused

Rumoured to be used by Willowsoft Overture File System (OFS1), if there is such a thing.

21 Reserved

(according to delorie). And Powerquest writes `Officially listed as reserved (HP Volume Expansion, SpeedStor variant)'. See also ID **a1**.)

21 Unused

Claimed for FSo2 (Oxygen File System) by Dave Poirier (`ekstazya@sprint.ca`). See dejanews.

22 Unused

Claimed for Oxygen Extended Partition Table by `ekstazya@sprint.ca`. See dejanews.

23 Reserved
24 NEC DOS 3.x

This is NEC MS-DOS 3.30 logically sectored FAT. Similar to type **14** above, the MBR could have up to 8 partition entries.

26 Reserved
27 PQservice

Acer laptop hidden rescue partition. Must be FAT32. Press Alt-F10 during boot to start this. Also other manufacturers use this type for their rescue partition.

27 Windows RE hidden partition

On MBR disks, type 0x27. On GPT disks, GUID: DE94BBA4-06D1-4D40-A16A-BFD50179D6AC. A hidden version of a Windows RE type 0x7 partition with NTFS. When this is installed, reboot and press F8 in order to boot into this Recovery Environment.

27 MirOS partition

MirOS BSD is a BSD variant.

27 RouterBOOT kernel partition

See RB500_Linux_SDK. RouterBOOT loads the contents of first partition with type 39 (0x27). If there is no such partition, it displays the message "CF BOOT FAILURE: kernel partition missing!". There is no filesystem on this partition, it contains a raw ELF Linux kernel image.

2a AtheOS File System (AFS)

AtheOS is an open source operating system written by Kurt Skauen. It is dead now - for a single page, see www.atheos.cx or sourceforge. For the history, see wikipedia. When progress seemed to stop, the project forked and the Syllable OS was started by Kristian van der Vliet (2002). See also wikipedia. It uses the same filesystem, AthFS or AFS, an extension of BeFS, the filesystem of BeOS. There is an attempt at a Linux driver at sourceforge.

2b SyllableSecure (SylStor)

A variation on AthFS is Sylstor, with added security.

31 Reserved
32 NOS

Simon Butcher (`simonb@alien.net.au`) writes: This type is being used by an operating system being developed by Alien Internet Services in Melbourne Australia called NOS. The id '32' was chosen not only because it's one of the few that are left available, but 32k is the size of the EEPROM the OS was originally targetted for.

33 Reserved
34 Reserved
35 JFS on OS/2 or eCS

David van Enckevort (`david@mensys.nl`) writes: *Type 0x35 is used by OS/2 Warp Server for e-Business, OS/2 Convenience Pack (aka version 4.5) and eComStation (eCS, an OEM version of OS/2 Convenience Pack) for the OS/2 implementation of JFS (IBM AIX Journaling Filesystem).* Since JFS is a non-bootable file system, you cannot install eCS to a JFS partition.

36 Reserved
38 THEOS ver 3.2 2gb partition
39 Plan 9 partition

Plan 9 is an operating system developed at Bell Labs for many architectures. Source is available. See comp.os.plan9. Originally Plan 9 used an unallocated portion at the end of the disk. Plan 9 3rd edition uses partitions of type 0x39, subdivided into subpartitions described in the Plan 9 partition table in the second sector of the partition.

39 THEOS ver 4 spanned partition
3a THEOS ver 4 4gb partition
3b THEOS ver 4 extended partition

THEOS is a multiuser multitasking OS for PCs founded by Timothy Williams in 1983. Current release 4.0, previous release 3.2. They say about themselves: `THEOS with over 150,000 customers and over 1,000,000 users around the world brings a mainframe look and feel to computers without the complexity and high maintenance costs. Hundreds of applications exist with networking and Windows integration.' See the Theos home page

3c PartitionMagic recovery partition

Cody Batt (codyb@powerquest.com) writes: When a PowerQuest product like PartitionMagic or Drive Image makes changes to the disk, it first changes the type flag to 0x3C so that the OS won't try to modify it etc. At the end of the process, it gets changed back to what it was at first. So, the only time you should see a 0x3C type flag is if the process was interrupted somehow (power outage, user reboot etc). If you change it back manually with a partition table editor or something then most of the time everything is okay.

3d Hidden NetWare

According to Powerquest.

40 Venix 80286

A very old Unix-like operating system for PCs.

40 PICK

Ross Stell writes: *The PICK multi-user operating system, developed in 1965 by Don Nelson and Dick Pick to run on an IBM 360 mainframe and implemented during the 1970s by many licensee companies on their minicomputers, was ported in 1983 by PICK Systems (http://www.marcomguy.com/pdf/CORP.PDF) to operate on the IBM PC-XT and later the AT and compatible PCs. This release is known as R83. Later (1989), PICK Systems produced Advanced PICK (AP) which operates within a Windows environment, thus obviating the need for a dedicated hard disk partition.* See also Wikipedia.

41 Linux/MINIX (sharing disk with DRDOS)

Very old FAQs recommended to use **41** etc instead of **81** etc on a disk shared with DRDOS because DRDOS allegedly disregards the high order bit of the partition type. (Or, rather, uses the high order bit to indicate that the partition is secured.) These types are not used anymore today. Roger Wolff (R.E.Wolff@BitWizard.nl) confirms: *I remember installing DRDOS, and getting a few extra drive letters that I didn't expect. Turns out those are my Minix partitions. It is looking at them as a FAT filesystem. Looks like a big mess. After finding no other possibility than to just "not touch those drive letters" I continue with the install. After a few minutes DRDOS automatically decides to write a copy of the FAT into a file on one of my MINIX partitions. Bye bye Minix partition.*

41 Personal RISC Boot

41 PPC PReP (Power PC Reference Platform) Boot
42 Linux swap (sharing disk with DRDOS)
42 SFS (Secure Filesystem)

SFS is an encrypted filesystem driver for DOS on 386+ PCs, written by Peter Gutmann.

42 Windows 2000 dynamic extended partition marker

If a partition table entry of type 0x42 is present in the legacy partition table, then W2K ignores the legacy partition table and uses a proprietary partition table and a proprietary partitioning scheme (LDM or DDM). As the Microsoft KnowledgeBase writes: *Pure dynamic disks (those not containing any hard-linked partitions) have only a single partition table entry (type **42**) to define the entire disk. Dynamic disks store their volume configuration in a database located in a 1-MB private region at the end of each dynamic disk.*

43 Linux native (sharing disk with DRDOS)
44 GoBack partition

GoBack is a utility that records changes made to the disk, allowing you to view or go back to some earlier state. It takes over disk I/O like a Disk Manager would, and stores its logs in its own partition.

45 Boot-US boot manager

Ulrich Straub (`ustraub@boot-us.de`) writes: The boot manager can be installed to MBR, a separate primary partition or diskette. When installed to a primary partition this partition gets the ID 45h. This partition does not contain a file system, it contains only the boot manager and occupies a single cylinder (below 8 GB). See www.boot-us.com.

45 Priam

According to Powerquest. See also ID 5c.

45 EUMEL/Elan
46 EUMEL/Elan
47 EUMEL/Elan
48 EUMEL/Elan

Eumel, and later Ergos L3, are multiuser multitasking systems developed by Jochen Liedtke at GMD. It was used at German schools for the computer science education. (Elan was the programming language used.)

4a Mark Aitchison's ALFS/THIN lightweight filesystem for DOS

According to Powerquest.

4a AdaOS Aquila (Withdrawn)

Nick Roberts at some point in time announced that he would use **4a** for Aquila, but now plans to use the AODPS **7f**.

4c Oberon partition

See http://www.oberon.ethz.ch/betadocu.html. This partition type (decimal 76) is used for the Aos filesystem. Type **4f** is used for the Nat filesystem. One may have several partitions of this type.

4d QNX4.x
4e QNX4.x 2nd part
4f QNX4.x 3rd part

QNX is a POSIX-certified, microkernel, distributed, fault-tolerant OS for the 386 and up, including support for the 386EX in embedded applications. For info see http://www.qnx.com/ or ftp.qnx.com. See also comp.os.qnx. ID 7 is outdated - QNX2 used **07**, QNX4.x uses decimal 77, and optionally 78 and 79 for additional QNX partitions on a single drive. See also **b1**-**b3** (decimal 177-179). See QNX Partitions and Neutrino filesystems.

4f Oberon partition

See http://www.oberon.ethz.ch/native/. (The partition ID is given in this posting in comp.lang.oberon. The install instructions say that at most one partition can have this type (decimal 79), and that one needs a different type, like **50** (decimal 80) for a second Oberon system. Moreover, that users of System Commander must avoid types containing the 0x10 bit.) See also type **4c** (decimal 76) above.

50 OnTrack Disk Manager (older versions) RO

Disk Manager is a program of OnTrack, to enable people to use IDE disks that are larger than 504MB under DOS. For info see http://www.ontrack.com. Linux kernel versions older than 1.3.14 do not coexist with DM.

50 Lynx RTOS

"Beginning with version 3.0, LynxOS gives users the ability to place up to 14 partitions of 2 GB each on both SCSI and IDE drives, for a total of up to 28 GB of file system space." See www.lynuxworks.com.

50 Native Oberon (alt)
51 OnTrack Disk Manager RW (DM6 Aux1)
51 Novell
52 CP/M
52 Microport SysV/AT
53 Disk Manager 6.0 Aux3
54 Disk Manager 6.0 Dynamic Drive Overlay (DDO)
55 EZ-Drive

EZ-Drive is another disk manager (by MicroHouse, 1992). Linux kernel versions older than 1.3.29 do not coexist with EZD. (On 990323 MicroHouse International was acquired by EarthWeb; MicroHouse Solutions split off and changed its name into StorageSoft. MicroHouse Development split off and changed its name into ImageCast. It is StorageSoft that now markets EZDrive and DrivePro.)

56 Golden Bow VFeature Partitioned Volume.

This is a Non-Standard DOS Volume. (Disk Manager type utility software)

56 DM converted to EZ-BIOS
56 AT&T MS-DOS 3.x logically sectored FAT.
57 DrivePro

Doug Anderson (`DougA@ImageCast.com`), with his brother Steve cofounder of MicroHouse (1989), writes: We actually use three different partition types: $55: `StorageSoft EZ-BIOS' - EZ-Drive, Maxtor, MaxBlast, and DriveGuide install this type if the drive needs to be handled by our INT13 redirector. $56: `StorageSoft EZ-BIOS DM Conversion' - Same as $55 but used when a DiskManager "skewed" partition has been converted to EZ-BIOS. $57: `StorageSoft DrivePro' - Used by our DrivePro product.

57 VNDI Partition

(According to `disk.c` in the Netware source. Not in actual use.)

5c Priam EDisk

Priam EDisk Partitioned Volume. This is a Non-Standard DOS Volume. (Disk Manager type utility software)

61 SpeedStor

Storage Dimensions SpeedStor Volume. This is a Non-Standard DOS Volume. (Disk Manager type utility software)

63 Unix System V (SCO, ISC Unix, UnixWare, ...), Mach, GNU Hurd

A Unixware 7.1 partition must start below the 4GB limit. (If the /stand/stage3.blm is located past this limit, booting will fail with "FATAL BOOT ERROR: Can't load stage3".)

64 PC-ARMOUR protected partition

Used by PC-ARMOUR, a disk protection by Dr. A.Solomon, intended to keep the disk inaccessible until the right password was given (and then an int13 hook was loaded above top-of-memory that showed c/h/s 0/0/2, with a copy of the real partition table, when 0/0/1 was requested). (`loekw@worldonline.nl`)

64 Novell Netware 286, 2.xx

65 Novell Netware 386, 3.xx or 4.xx

(Novell Netware used to be the main Network Operating System available. Netware 68 or S-Net (1983) was for a Motorola 68000, Netware 86 for an Intel 8086 or 8088. Netware 286 was for an Intel 80286 and existed in various versions that were later merged to Netware 2.2. Netware 386 was a rewrite in C for the Intel 386, later renamed 3.x - it existed at least in versions 3.0, 3.1, 3.10, 3.11, 3.12. Its successor Netware 4.xx had versions 4.00, 4.01, 4.02, 4.10, 4.11. Then came Intranetware.) Netware >= 3.0 uses one partition per drive. It allocates logical Volumes inside these partitions. The volumes can be split over several drives. The filesystem used is called "Turbo FAT"; it only very vaguely resembles the DOS FAT file system. (Kai Henningsen (`kai@khms.westfalen.de`))

66 Novell Netware SMS Partition

According to `disk.c` in the Netware source. SMS: Storage Management Services. No longer used.

67 Novell

Roman Gruber reports: this code has frozen my version of norton disk-editor (so I think it has to be something special). Jeff Merkey says: 67 is for Wolf Mountain.

68 Novell
69 Novell Netware 5+, Novell Netware NSS Partition

According to `disk.c` in the Netware source. NSS = Novell Storage Services.

6e ??

Reported once.

70 DiskSecure Multi-Boot
71 Reserved
72 V7/x86

Robert Nordier writes: V7/x86, a port of UNIX Version 7 to the PC, is available at www.nordier.com/v7x86.

73 Reserved
74 Reserved
74 Scramdisk partition

Scramdisk is freeware and shareware disk encryption software. It supports container files, dedicated partitions (type 0x74) and disks hidden in WAV audio files. (Shaun Hollingworth (`moatlane@btconnect.com`))

75 IBM PC/IX
76 Reserved
77 M2FS/M2CS partition

Jeff Merkey writes: 77 is one we are using internally for M2FS/M2CS partitions.

77 VNDI Partition

(According to `disk.c` in the Netware source. Not in actual use.)

78 XOSL FS

XOSL Bootloader filesystem, see xosl2.com.

7e Unused

Claimed for F.I.X. by `gruberr@kapsch.net`. See dejanews.

7f Unused

Proposed for the Alt-OS-Development Partition Standard.

80 MINIX until 1.4a
81 MINIX since 1.4b, early Linux

Minix is a Unix-like operating system written by Andy Tanenbaum and students at the Vrije Universiteit, Amsterdam, around 1989-1991. It runs on PCs (8086 and up), MacIntosh, Atari, Amiga, Sparc. Ref: Operating Systems: Design and Implementation, Andrew S. Tanenbaum, Prentice-Hall, ISBN 0-13-637406-9 Since 950601 Minix is freely available - site: ftp.cs.vu.nl. See also comp.os.minix.

81 Mitac disk manager
82 Prime
82 Solaris x86

Solaris creates a single partition with id 0x82, then uses Sun disk labels within the partition to split it further. (Brandon S. Allbery (`allbery@kf8nh.apk.net`)) Starting from 2005, newly installed systems will use 0xbf.

82 Linux swap
83 Linux native partition

Linux is a Unix-like operating system written by Linus Torvalds and many others on the internet since Fall 1991. It runs on PCs (386 and up) and a variety of other hardware. It is distributed under GPL. Software can be found numerous places, like ftp.funet.fi, metalab.unc.edu and tsx-11.mit.edu. See also comp.os.linux.* and http://www.linux.org/. Various filesystem types like xiafs, ext2, ext3, reiserfs, etc. all use ID 83. Some systems mistakenly assume that 83 must mean ext2.

84 OS/2 hidden C: drive

OS/2-renumbered type **04** partition.

84 Hibernation partition

(following Appendix E of the Microsoft APM 1.1f specification). Reported for various laptop models. E.g., used on Dell Latitudes (with Dell BIOS) that use the MKS2D utility. APM 1.2 hibernation partitions can be used by Windows 98 or higher.

85 Linux extended partition
86 Old Linux RAID partition superblock

See **fd**.

86 FAT16 volume set

Legacy Fault Tolerant FAT16 volume. Windows NT 4.0 or earlier will add 0x80 to the partition type for partitions that are part of a Fault Tolerant set (mirrored or in a RAID-5 volume). Thus, one gets types **86**, **87**, **8b**, **8c**. See also Windows NT Boot Process and Hard Disk Constraints.

87 NTFS volume set

Legacy Fault Tolerant NTFS volume. HPFS Fault-Tolerant mirrored partition.

88 Linux plaintext partition table
8a Linux Kernel Partition (used by AiR-BOOT)

Martin Kiewitz (`KiWi@vision.fido.de`) writes: I'm currently writing a pretty nice boot-loader. For this I'm using Linux Boot Loader ID A0h, and partitition type 8Ah for the partition holding the kernel image.

8b Legacy Fault Tolerant FAT32 volume
8c Legacy Fault Tolerant FAT32 volume using BIOS extd INT 13h
8d Free FDISK 0.96+ hidden Primary DOS FAT12 partitition

Free FDISK is the FDISK used by FreeDOS. It hides types **01**, **04**, **05**, **06**, **0b**, **0c**, **0e**, **0f** by adding decimal 140 (0x8c).

8e Linux Logical Volume Manager partition

See pvcreate(8) as found under http://linux.msede.com/lvm. (For a while this was 0xfe.)

90 Free FDISK 0.96+ hidden Primary DOS FAT16 partitition
91 Free FDISK 0.96+ hidden DOS extended partitition
92 Free FDISK 0.96+ hidden Primary DOS large FAT16 partitition
93 Hidden Linux native partition
93 Amoeba
94 Amoeba bad block table

Amoeba is a distributed operating system written by Andy Tanenbaum, together with Frans Kaashoek, Sape Mullender, Robert van Renesse and others since 1981. It runs

on PCs (386 and up), Sun3, Sparc, 68030. It is free for universities for research/teaching purposes. For information, see ftp.cs.vu.nl.

95 MIT EXOPC native partitions

http://www.pdos.lcs.mit.edu/exo/ (Andrew Purtell, Andrew_Purtell@NAI.com)

96 CHRP ISO-9660 filesystem
97 Free FDISK 0.96+ hidden Primary DOS FAT32 partitition
98 Free FDISK 0.96+ hidden Primary DOS FAT32 partitition (LBA)
98 Datalight ROM-DOS Super-Boot Partition

See www.datalight.com, and type **12** above.

99 DCE376 logical drive

No, it's not a hibernation partition; it's closest to a DOS extended partition. It's used by the Mylex DCE376 EISA SCSI adaptor for partitions which are beyond the 1024th cylinder of a drive. I've only seen references to type **99** with the DCE376. (Christian Carey, ccarey@CapAccess.ORG)

9a Free FDISK 0.96+ hidden Primary DOS FAT16 partitition (LBA)
9b Free FDISK 0.96+ hidden DOS extended partitition (LBA)
9e ForthOS partition

ForthOS is the name Andy Valencia uses for his Forth operating system, a port of eForth. Also the older VSTA by the same author uses partition type 0x9e (158).

9f BSD/OS

Current sysid for BSDI. The types **b7** and **b8** given below are for an older version of the filesystem used in pre-v3.0 versions of the OS. These days the system is v4.1 BSD/OS. BSDI reports 2.1 million installed servers and 12 million licenses sold. See http://www.bsdi.com/.

a0 Laptop hibernation partition

Reported for various laptops like IBM Thinkpad, Phoenix NoteBIOS, Toshiba under names like zero-volt suspend partition, suspend-to-disk partition, save-to-disk partition, power-management partition, hibernation partition. Usually at the start or end of the disk area. (This is also the number used by Sony on the VAIO. Recent VAIOs can also hibernate to a file in the filesystem, the choice being made from the BIOS setup screen.)

a1 Laptop hibernation partition

Reportedly used as "Save-to-Disk" partition on a NEC 6000H notebook. Types **a0** and **a1** are used on systems with Phoenix BIOS; the Phoenix PHDISK utility is used with these.

a1 HP Volume Expansion (SpeedStor variant)

IDs 21, a1, a3, a4, a6, b1, b3, b4, b6 are for HP Volume Expansion (SpeedStor variant).

a3 HP Volume Expansion (SpeedStor variant)
a4 HP Volume Expansion (SpeedStor variant)
a5 BSD/386, 386BSD, NetBSD, FreeBSD

386BSD is a Unix-like operating system, a port of 4.3BSD Net/2 to the PC done by Bill Jolitz around 1991. When Jolitz seemed to stop development, an updated version was called FreeBSD (1992). The outcome of a Novell vs. UCB law suit was that Net/2 contained AT&T code, and hence was not free, but that 4.4BSD-Lite was free. After that, FreeBSD and NetBSD were restructured, and FreeBSD 2.0 and NetBSD 1.0 are based on 4.4BSD-Lite. FreeBSD runs on PCs. See http://www.freebsd.org/FreeBSD.html. For NetBSD, see below - it changed partition type to **a9**. 386BSD seems to be dead now. The kernel source is being published - see Operating System Source Code Secrets by Bill and Lynne Jolitz. See comp.os.386bsd.*. See http://www.paranoia.com/~vax/boot.html for NetBSD boot and partitioning info.

a6 OpenBSD

OpenBSD, led by Theo de Raadt, split off from NetBSD. It tries to emphasize on security. See http://www.openbsd.org/.

a6 HP Volume Expansion (SpeedStor variant)
a7 NeXTStep

Based on Mach 2.6 and features of Mach 3.0, is a true object-oriented operating system and user environment. See http://www.next.com/.

a8 Mac OS-X

Apple's OS-X (Darwin Intel) uses this type for its filesystem partition (a UFS file system, in NeXT flavour, only differing from the *BSD formats in the first 8 KB). See also type **ab**.

a9 NetBSD

NetBSD is one of the children of *BSD (see above). It runs on PCs and a variety of other hardware. Since 19-Feb-98 NetBSD uses **a9** instead of **a5**. See http://www.netbsd.org/. It is freely obtainable - see http://www.netbsd.org/Sites/net.html.

aa Olivetti Fat 12 1.44MB Service Partition

Contains a bare DOS 6.22 and a utility to exchange types **06** and **aa** in the partition table. (`loekw@worldonline.nl`)

ab Mac OS-X Boot partition

Apple's OS-X (Darwin Intel) uses this type for its boot partition. The image (`/usr/standalone/i386/boot`) starts at sector 1. See also type **a8**.

ab GO! partition

Unused. Claimed by Stanislav Karchebny for his GO! OS.

ae ShagOS filesystem
af ShagOS swap partition

Unused. Claimed by Frank Barrus for his ShagOS.

af MacOS X HFS

Used by Apple for the MacOS X filesystem HFS or HFS+ on Intel.

b0 BootStar Dummy

The boot manager BootStar manages its own partition table, with up to 15 primary partitions. It fills unused entries in the MBR with BootStar Dummy values. See www.star-tools.com. If you use this, don't use a disk manager, do not put LILO in the MBR and do not use fdisk.

b1 HP Volume Expansion (SpeedStor variant)
b1 QNX Neutrino Power-Safe filesystem
b2 QNX Neutrino Power-Safe filesystem
b3 HP Volume Expansion (SpeedStor variant)
b3 QNX Neutrino Power-Safe filesystem

Steve Reid (`stever@qnx.com`) writes: We've recently added a new Power-Safe filesystem to the QNX Neutrino RTOS, and it uses decimal 177, 178, and 179 (B1, B2, and B3) for its identifier. See QNX6 fs. Default is 179.

b4 HP Volume Expansion (SpeedStor variant)
b6 HP Volume Expansion (SpeedStor variant)
b6 Corrupted Windows NT mirror set (master), FAT16 file system
b7 Corrupted Windows NT mirror set (master), NTFS file system
b7 BSDI BSD/386 filesystem
b8 BSDI BSD/386 swap partition

BSDI (Berkeley Software Design, Inc.) was founded by former CSRG (UCB Computer Systems Research Group) members. Their operating system, based on Net/2, was called BSD/386. After the USL (Unix System Laboratories, Inc./Novell Corp.) vs. BSDI lawsuit, new releases were based on BSD4.4-Lite. Now they are announcing BSD/OS V2.0.1. This is an operating for PCs (386 and up), boasting 3000 customers. (That was long ago. The current partition id is **9f**, see above.)

bb Boot Wizard hidden

(PTS) BootWizard 4.0 and its new version Acronis OS Selector 5.0 use this id (i) when hiding partitions with types other than **01**, **04**, **06**, **07**, **0b**, **0c**, **0e**, and (ii) when creating a partition without file system. See www.PhysTechSoft.com. The boot software was purchased on 2001-01-05 by SWsoft. See www.acronis.com.

bc Acronis backup partition

Recognized as Acronis Secure Zone, when labelled "ACRONIS SZ". A primary partition, formatted with FAT32, LBA mapped.

be Solaris 8 boot partition
bf New Solaris x86 partition

The old 0x82 id conflicted with Linux swap. New Solaris installations will use the id 0xbf. (Larry Lee <lclee@west.sun.com>)

c0 CTOS
c0 REAL/32 secure small partition

See d0 below.

c0 NTFT Partition

According to `disk.c` in the Netware source.

c0 DR-DOS/Novell DOS secured partition

DR-DOS 7.02+ / OpenDOS 7.01 / Novell DOS 7 secured partition.

c1 DRDOS/secured (FAT-12)
c2 Unused

According to Powerquest IDs **c2**, **c3**, **c8**, **c9**, **ca**, **cd** are reserved for DR-DOS 7+. According to Matthias Paul **c2**, **c3**, **cd** are no longer reserved for DR-DOS.

c2 Hidden Linux
c3 Hidden Linux swap

Benedict Chong (bchong@blueskyinnovations.com) writes: BlueSky Innovations LLC does a boot manager product called Power Boot and we use, in addition, 0C2h and 0C3h for hidden Linux partitions (swap and ext2fs). See also ID c2.

c4 DRDOS/secured (FAT-16, < 32M)
c5 DRDOS/secured (extended)

This ID may also be used in obscure trickery: on a shared MS-DOS / DR-DOS machine with DR-DOS 6.0-7.03 (so that the DR_DOS does not understand type **0f** and the MS-DOS does not understand type **c5**) one may have two extended partitions, where each operating system sees only one.

c6 DRDOS/secured (FAT-16, >= 32M)

DR-DOS 6.0 and higher (NetWare PalmDOS 1.0, Novell DOS 7, OpenDOS 7.01, DR-DOS 7.02+) will add 0xc0 to the partition type for a LOGIN.EXE-secured partition (so that people cannot avoid the password check by booting from an MS-DOS floppy). Otherwise it seems that the types **c1**, **c4**, **c5**, **c6** and **d1**, **d4**, **d5**, **d6** are used precisely like **01**, **04**, **05**, **06** (but are accepted only when booting from disk).

c6 Windows NT corrupted FAT16 volume/stripe set

NTFS will add 0xc0 to the partition type for disabled parts of a Fault Tolerant set. Thus, one gets types **c6**, **c7**. See also Windows NT Boot Process and Hard Disk Constraints and Switching from DR-DOS 6.0 to MS-DOS 5.0.

c7 Windows NT corrupted NTFS volume/stripe set
c7 Syrinx boot

Primary partition only.

c8 Reserved for DR-DOS 8.0+
c9 Reserved for DR-DOS 8.0+
ca Reserved for DR-DOS 8.0+
cb DR-DOS 7.04+ secured FAT32 (CHS)/
cc DR-DOS 7.04+ secured FAT32 (LBA)/
cd CTOS Memdump?
ce DR-DOS 7.04+ FAT16X (LBA)/
cf DR-DOS 7.04+ secured EXT DOS (LBA)/
d0 REAL/32 secure big partition

REAL/32 is a continuation of DR Multiuser DOS. It supports FAT12, FAT16 and REAL/32 7.90 also supports FAT32. Andrew Freeman (`afreeman@imsltd.com`) writes: For partitions which have been marked as secure we use 0xC0 and 0xD0 as partition markers (C0 < 32mb, D0 >= 32mb). REAL/32 is an advanced 32-bit multitasking & multi-user MS-DOS & Windows compatible operating system. Home page is www.imsltd.com.

d0 Multiuser DOS secured partition

This applies to the whole MDOS family range, Digital Research DR Multiuser DOS and Novell DR Multiuser DOS, as well as to Concurrent Controls Multiuser DOS, Datapaq Australasia System Manager 7, and IMS Multiuser DOS.

d1 Old Multiuser DOS secured FAT12
d4 Old Multiuser DOS secured FAT16 <32M
d5 Old Multiuser DOS secured extended partition
d6 Old Multiuser DOS secured FAT16 >=32M
d8 CP/M-86
da Non-FS Data

Added on request of John Hardin (`johnh@aproposretail.com`).

da Powercopy Backup

Powercopy Backup (shielded disk), used by www.datapower.de.

db Digital Research CP/M, Concurrent CP/M, Concurrent DOS
db CTOS (Convergent Technologies OS -Unisys)
db KDG Telemetry SCPU boot

Mark Morgan Lloyd (markMLl.in@telemetry.co.uk) writes: KDG Telemetry uses type 0xdb to store a protected-mode binary image of the code to be run on a 'x86-based SCPU (Supervisory CPU) module from the DT800 range.

dd Hidden CTOS Memdump?
de Dell PowerEdge Server utilities (FAT fs)
df DG/UX virtual disk manager partition

Glenn Steen (glenn.steen@ap1.se) writes: When I made an old Aviion 2000 triple-boot (DOS, DG/UX and Linux) I saw that Linux fdisk reported the DG/UX virtual disk manager partition as type 0xdf.

df BootIt EMBRM

The boot manager BootIt manages its own partition table, with up to 255 primary partitions. See www.terabyteunlimited.com. If you use this, don't use a disk manager, do not put LILO in the MBR and do not use fdisk. Reference for the ID: BOOTIT.TXT.

e0 Reserved by STMicroelectronics for a filesystem called ST AVFS.
e1 DOS access or SpeedStor 12-bit FAT extended partition

Kevin Cummings reports in alt.os.linux: it's a SSTOR partition on cylinders > 1023.

e3 DOS R/O or SpeedStor
e4 SpeedStor 16-bit FAT extended partition < 1024 cyl.
e5 Tandy MSDOS with logically sectored FAT
e6 Storage Dimensions SpeedStor
e8 LUKS

LUKS (Linux Unified Key Setup) partition.

eb BeOS BFS

BeOS is an operating system that runs on Power PCs and on Intel PCs. Version 5 (the last version) is distributed freely to individuals. The system was sold to Palm and is not developed any more. OpenBeOS tries to create an open source version.

ec SkyOS SkyFS

SkyOS is an operating system written by Robert Szeleney. Its filesystem SkyFS is based on OpenBeFS.

ed Unused

Matthias Paul plans to use this for an OS called Sprytix.

ee Indication that this legacy MBR is followed by an EFI header
ef Partition that contains an EFI file system

Bob Griswold (rogris@Exchange.Microsoft.com) writes: MS plans on using EE and EF in the future for support of non-legacy BIOS booting. Mark Doran (mark.doran@intel.com) adds: these types are used to support the Extensible Firmware Interface specification (EFI); go to developer.intel.com and search for EFI. (For the types **ee** and **ef**, see Tables 16-6 and 16-7 of the EFI specification, EFISpec_091.pdf.)

f0 Linux/PA-RISC boot loader

Paul Bame (bame@debian.org) writes: the F0 partition will be located in the first 2GB of a drive and used to store the Linux/PA-RISC boot loader and boot command line, optionally including a kernel and ramdisk.

f1 Storage Dimensions SpeedStor
f2 DOS 3.3+ secondary partition

Matthias Paul writes: "This ID was originally used by Sperry IT MS-DOS 3.xx for a logically sectored variant of FAT. When Sperry IT became part of Unisys, the operating system was called Unisys MS-DOS 3.3. Digital Research's DOS Plus 2.1 (for OEM machines such as the Amstrad/Schneider PC1512, the T.R.A.N. Jasmin Turbo (Speed 8M), or the Acorn BBC Master 512 also supports this ID and logs it in, as if this would be either a type 01h FAT12 or a type 04h FAT16 partition."

f3 Reserved

Powerquest writes: Storage Dimensions SpeedStor.

f4 SpeedStor large partition
f4 Prologue single-volume partition
f5 Prologue multi-volume partition

The type F4 partition contains one volume, and is not used anymore. The type F5 partition contains 1 to 10 volumes (called MD0 to MD9). It supports one or more systems (Prologue 3, 4, 5, Twin Server). Each volume can have as file system the NGF file system or TwinFS file system. NGF (old): volume size at most 512 MB, at most 895 files per directory, at most 256 directories per volume. TwinFS (new): volume size up to 4 GB. No limit in number of files and directories. See Prologue.

f6 Storage Dimensions SpeedStor
f7 Unused

Maybe Natalia Portillo plans to use this for O.S.G. EFAT ("Enhanced File Allocation Techniques").

f9 pCache

Ed Sawicki writes: "We propose using the F9 partition type as a pCache partition, which is our name for an "ext2/ext3 persistent cache partition". See www.alcpress.com.

fa Bochs

Rob Judd writes: MandrakeSoft's Bochs x86 emulator (similar to VMWare) uses **fa** as a partition identifier.

fb VMware File System partition
fc VMware Swap partition

Also used as VMkernel dump partition. (Cf. vmwareguide.)

VMware offers virtual machines in which one can run Linux, Windows, FreeBSD.

fd Linux raid partition with autodetect using persistent superblock

See the HOWTO and the kernel patches. Earlier, **86** was used instead of **fd**.

Powerquest writes: Reserved for FreeDOS (www.freedos.org), but it seems FreeDOS never used this ID.

fe SpeedStor > 1024 cyl.
fe LANstep
fe IBM PS/2 IML (Initial Microcode Load) partition, located at the end of the disk.
fe Windows NT Disk Administrator hidden partition

Mark Morgan Lloyd (markMLl.in@telemetry.co.uk) writes: Windows NT Disk Administrator marks hidden partitions (i.e. present but not to be accessed) as type 0xfe. A primary partition of this type is also used by IBM to hold an image of the "Reference Diskettes" on many of their machines, particularly newer PS/2 systems (at a rough guess, anything built after about 1994). This clash can cause major confusion and grief if running NT on IBM kit. When this Reference Partition is activated, it changes its type into 1 (FAT12) and hides all other partitions by adding 0x10 to the type.

fe Linux Logical Volume Manager partition (old)

This has been in use since the early LVM days back in 1997, and has now (Sept. 1999) been renamed 0x8e.

ff Xenix Bad Block Table
